HOMEWORK 1: 1.1-2.6

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1.1. Let $x = (1,2,3)^T$, $y = (y_1,y_2,y_3)^T$, $z = (4,2,1)^T$. Compute $2x$, $3y$, and $x+2y-3z$.

Proof. 1: $2x = 2(1,2,3)^T = (2,4,6)^T$
2: $3y = 3(y_1,y_2,y_3)^T = (3y_1,3y_2,3y_3)^T$
3: $x+2y-3z = (1,2,3)^T+2(y_1,y_2,y_3)^T-(12,6,3)^T = (2y_1-11,2y_2-4,2y_3)^T$ □

1.2. Which of the following sets (with natural addition and multiplcation by a scalar) are vector spaces? Justify your answer.

a) The set of all continuous functions on the interval $[0,1]$;
b) The set of all non-negative functions on the interval $[0,1]$;
c) The set of all polynomials of degree exactly n;
d) The set of all symmetric $n \times n$ matrices, i.e the set of matrices $A = \{a_{j,k}\}_{j,k=1}^n$ such that $A^T = A$.

Proof. A is a vector space. Given two distinct functions f and g, and some $x \in [0,1]$, we can define vector addition and scalar multiplication as such: $(f+g)(x) = f(x)+g(x)$; $(\alpha f)(x) = \alpha f(x)$, where $\alpha \in \mathbb{R}$.

□

1.3. True or false:

a) Every vector space contains a zero vector;
b) A vector space can have more than one zero vector;
c) An $m \times n$ matrix has m rows and n columns;
d) If f and g are polynomials of degree n, then $f+g$ is also a polynomial of degree n;
e) If f and g are polynomials of degree at most n, then $f+g$ is also a polynomial of degree at most n

Proof. TFTFT □

1.4. Prove that a zero vector 0 of a vector space V is unique.

Proof. To prove uniqueness, assume that there exists two zero vectors, 0 and $\bar{0}$, both belonging to the vector space V. By our axioms, we know if $x \in V$, then $x+0 = x+\bar{0} = x$. Now, using our axioms, we can derive that $x+\bar{0} = x+0 \to \bar{0} = 0$. This shows that our distinct zero vectors are equal, and thus the zero vector is unique. □

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1.5. What matrix is the zero vector of the space $M_{2\times 3}$

Proof. $\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ □

1.6. Prove that the additive inverse, defined in Axiom 4 of a vector space is unique.

Proof. To prove uniqueness, assume there exists two additive inverses in V, that being $-v$ and $\bar{v}$. Using our axioms, we may derive that $v + (-v) = v + (\bar{v}) = 0$. Now, using our axioms, we can derive that $v + (-v) = v + (\bar{v}) \to -v = \bar{v}$. This shows that the additive inverse is unique. □

1.7. Prove that $0v = 0$ for any vector $v \in V$.

Proof. $0v = 0 \to (0 + 0)v = 0v \to 0v + 0v = 0v \to 0v = 0$. □

1.8. Prove that for any vector v its additive inverse $-v$ is given by $(-1)v$.

Proof. We know that $v + (-v) = 0$ through our axioms. Now, we can represent this using the distributive property: $(1 + (-1))v = 0 \to 0v = 0$ which is proven by our axiom for vector spaces. □

2.1. Find a basis in the space of 3×2 matrices $M_{3\times 2}$.

Proof. Since we have a 3×2 matrix, our basis for this vector space would be the first two columns of the matrix. Since we only have two columns, we span $\mathbb{R}^{\not 2}$ in $\mathbb{R}^3$ plane. A simpler way to put it is that we can only get two pivots in the matrix using RREF, thus the original columns would be the basis for the vector space. □

2.2. True or false:

a) Any set containing the zero vector is linearly dependent;
b) A basis must contain 0;
c) subsets of linearly dependent sets are linearly dependent;
d) subsets of linearly independent sets are linearly independent;
e) If $\alpha_1 v_1 + \alpha_2 v_2 + \ldots + \alpha_n v_n = 0$, then all scalars α_k are zero;

2.3. Recall, that a matrix is called symmetric if $A^T = A$. Write down a basis in the space of symmetric 2×2 matrices. How many elements are in the basis?

Proof. $\begin{pmatrix} a & b \\ b & c \end{pmatrix}$ is the format for these symmetric matrices. Now, using a linear combination to represent this: $a\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + b\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + c\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$ The vectors next to the coefficients are the elements of the basis. □

2.4. Write down a basis for the space of

a) 3×3 symmetric matrices;
b) $n \times n$ symmetric matrices;
c) $n \times n$ antisymmetric ($A^T = -A$) matrices;

2.5. Let a system of vectors $v_1, v_2, \ldots, v_r$ be linearly independent but not generating. Show that it is possible to find a vector v_{r+1} such that the system $v_1, v_2, \ldots, v_r, v_{r+1}$ is linearly independent.

Proof. Let's suppose that there is a vector v_{r+1} that can be represented as the linear combination $\sum_{k=1}^{r} \alpha_k v_k$. Set up: $\alpha_{r+1} v_{r+1} =$ □

2.6. Is it possible that vectors v_1, v_2, v_3 are linearly dependent, but the vectors $w_1 = v_1 + v_2$, $w_2 = v_2 + v_3$, $w_3 = v_3 + v_1$ are linearly independent?